Two circle fits on one admittance locus

Quantitative comparison of the two Butterworth-Van Dyke circle estimates openQCM NEXT computes from the same measured admittance: the overlay in the main window's impedance panel and the fit in the live admittance-fit window.

Branch impedance-analysis. Archived five-overtone air sweep, 2026-08-28. Every number reproduced by compare_circle_fits.py.

Summary

The two circles differ by 0.9 % to 10.1 % in radius, hence 0.9 % to 11.3 % in the implied R1, with a sign that never changes: the panel's circle is always the smaller one.

The cause is **which points are fitted**, not which algorithm fits them. Restricting the fit to abs(f - f_s) <= 1 half width reproduces almost the entire difference on its own; swapping the estimator while keeping the whole band accounts for at most a third of it, and on some overtones for nothing.

That selection matters at all only because the measured locus is **not a circle**. Its radial residual against the best-fit circle is 1.5 % to 4.3 % of the radius and is smooth and systematic, not scatter. The disagreement scales with that residual across the five overtones: R2 = 0.92, slope 3.0.

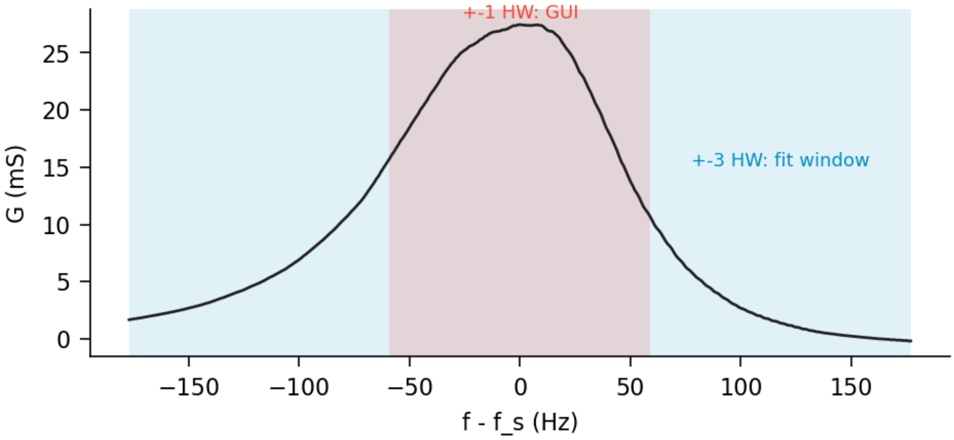
The quantity that separates the two views is therefore a model-error diagnostic, not a numerical accident. Where the BVD circle describes the data (n = 5, residual 1.5 %) the two agree to 0.9 %; where it does not (n = 1, residual 4.3 %) they diverge by 10 %. Neither estimate reaches the datalog: both views are display-only.

1 The two estimators

Both read the same three buffers - `get_G_exact_buffer`, `get_B_exact_buffer`, `get_F_G_values_buffer` - populated by MultiscanProcess from the exact complex inversion, and both draw a dashed circle over the measured locus.

	main-window panel	live fit window
implementation	MainWindow._fit_circle_taubin (ui/mainWindow.py)	fit1_circle (sweep_data/fit_admittance.py)
domain	decimated to ~250 samples, then abs(f - f_s) <= 1 half width	decimated to 250 samples, whole published band, +- 3 half widths
cost function	Taubin algebraic, then up to 6 rounds of 2-sigma trimming	Taubin as initial guess, then orthogonal-distance least squares
derived output	none; the circle is the result	theta, f_s, Gamma from the arc, R1, L1, C1, residual

The producer publishes +- 3 half widths, so the two domains differ by a factor of three in span and roughly a factor of three in sample count. Measured against the reference centre, the band subtends 293-304 degrees and the core 176-187 degrees.



Conductance of the fundamental. Blue: the published band, and the whole of what the fit window fits. Red: the core the panel restricts itself to.

2 Magnitude of the disagreement

Both estimators started from the identical array. R1 = 1 / 2r, so a radius short by 10 % is a resistance high by 11 %.

n	r ref	r panel	dr	R1 ref	R1 panel	dR1	band	core
	mS	mS	%	ohm	ohm	%	pts	pts
1	13.385	12.029	-10.1	37.36	41.56	+11.3	355	119
3	20.025	18.868	-5.8	24.97	26.50	+6.1	255	85
5	9.407	9.320	-0.9	53.15	53.65	+0.9	387	129
7	6.250	6.077	-2.8	80.00	82.27	+2.8	259	87
9	3.855	3.708	-3.8	129.71	134.84	+4.0	251	84

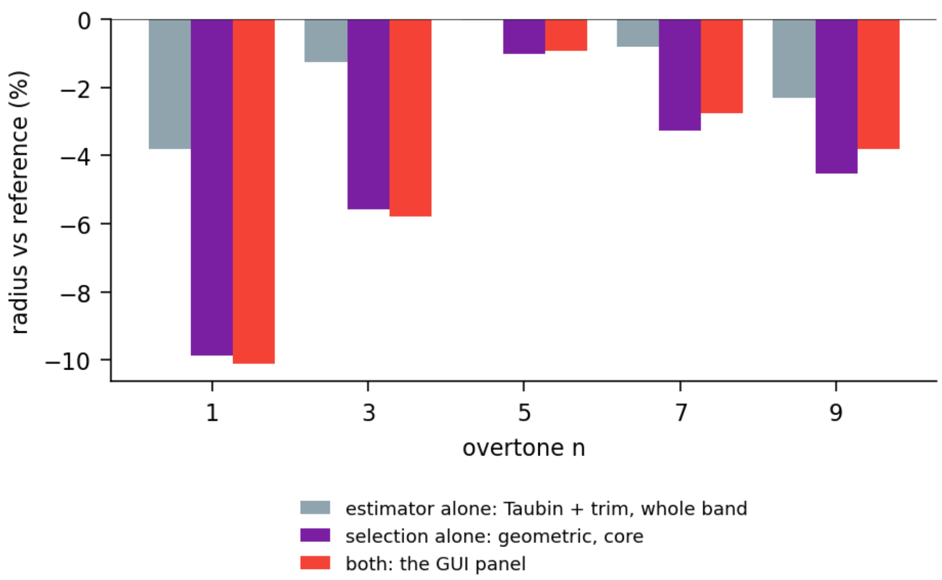
The centre moves consistently as well: on the fundamental from (12.761, -1.445) mS to (14.824, -1.904) mS, towards higher conductance, along the direction of the retained arc.

3 Decomposition: domain against estimator

The two views differ in two respects at once. Running the four combinations separates them. Reference is the geometric fit on the whole band; values are radius relative to it.

n	Taubin + trim, whole band	geometric, core	Taubin + trim, core (the panel)
1	-3.8 %	-9.9 %	-10.1 %
3	-1.3 %	-5.6 %	-5.8 %
5	-0.0 %	-1.0 %	-0.9 %
7	-0.8 %	-3.3 %	-2.8 %
9	-2.3 %	-4.5 %	-3.8 %

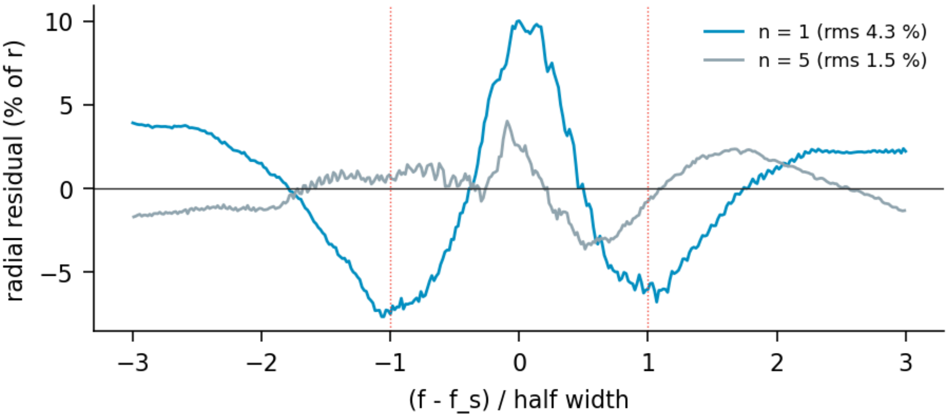
The domain dominates. Keeping the geometric estimator and restricting it to the core already yields -9.9 % of the -10.1 % measured on the fundamental; the estimator alone, on the full band, yields -3.8 %. Once the domain is the core, which estimator runs on it is close to irrelevant. The effects are not additive, consistent with both being expressions of one underlying cause rather than two independent errors.



Radius relative to the reference fit, by variant and overtone.

4 Why the domain matters: the locus is not a circle

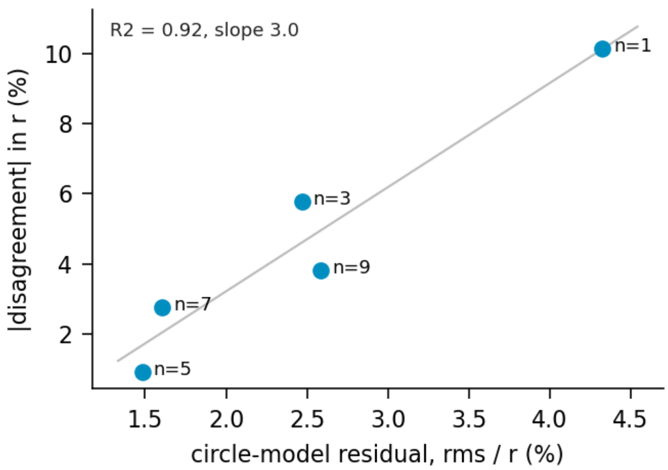
If the data lay on a circle, any consistent estimator on any subset containing three non-collinear points would return the same circle, and the table above would be zero throughout. It is not, so the model is incomplete. The residual quantifies by how much.



Signed radial residual against the best-fit circle, as a percentage of the radius, versus frequency in half widths. Dotted lines mark the core boundary.

The residual is a smooth function of frequency, not noise: on the fundamental it is +10 % of the radius at resonance and -7 % near +-1 half width, changing sign three times across the band. Its turning points sit at the core boundary, which is precisely why the choice of domain is consequential. A circle constrained to pass through an arc that bulges outward in its middle and inward at its ends settles at a smaller radius; a circle fitted to the whole band averages the excursion out.

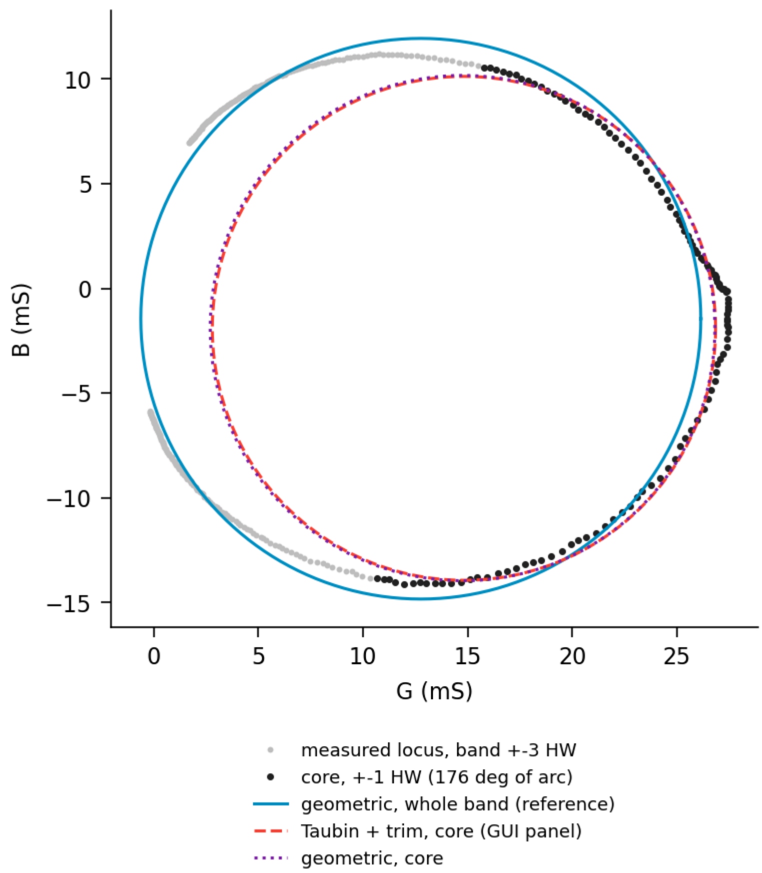
Across the five overtones the disagreement tracks the residual.



Disagreement in radius against the circle-model residual, one point per overtone.

R2 = 0.92, Pearson 0.961, Spearman 0.900, slope 3.0 - each percentage point of circle-model residual buys about three percentage points of radius disagreement. With five points this is an association, not a law, but the ordering is unambiguous and the mechanism is visible directly in the residual.

The residual also settles which fit is preferable on its own terms. Over the whole band the reference circle leaves 4.33 % rms on the fundamental and the panel's circle 17.91 %. Over the core only the ordering reverses, 5.88 % against 2.95 %, as it must, since that is the domain the panel optimised. Each estimator wins where it was fitted: the signature of model error, not of one estimator being numerically better.



Fundamental. Grey: the published band. Black: the core. Blue: geometric fit on the whole band. Red: the panel. Purple: geometric fit on the core, which almost coincides with the panel and confirms the domain as the cause.

5 What this affects

Within the fit window the radius enters R1 = 1 / 2r and through it L1 and C1. f_s and Gamma are read from the arc - from the centre and the rotation theta, not the radius - so they are affected only through the centre displacement, and second-order in it.

The panel reports no numbers at all; it draws a circle. The overlay is therefore the entire visible consequence, and the practical rule is that it must not be read as an estimate of R1.

Neither path reaches the datalog: _update_impedance_panel is display-only, and the logged frequency and dissipation still come from the approximate formula in MultiscanProcess. A disagreement of 10 % between the two overlays leaves the logged record untouched.

The fit window is the reference of the two, by construction rather than by argument: ui/impedanceFitWindow.py imports sweep_data/fit_admittance.py by file path, so the live figures and the offline script cannot diverge. Keeping both is deliberate: two estimates whose difference measures model error are more informative than one number that looks authoritative.

6 Method

```
cd software
QT_QPA_PLATFORM=offscreen python3 \
  ../research/admittance-circle-fit/compare_circle_fits.py <dir>
```

<dir> holds g1.txt ... g9.txt as written by the sweep dump. Copy them out of the repository before use: software/openQCM/sweep_data/ is overwritten by every acquisition.

The script rebuilds the admittance with the offline admittance(), applies the AD8302 ratio mask and the +- 3 half-width clip that MultiscanProcess applies, then runs both estimators on that single array - the same array both views receive at run time. Qt is required only because the panel's estimator is a static method on MainWindow; nothing is displayed. The decomposition of section 3 and the residuals of section 4 use the same preparation, with geometric_circle called directly for the mixed cases.

7 Limits

One instrument, one sweep, air. The residual is expected to grow in liquid, where damping widens the resonance and pushes more of the band into the AD8302 dynamic-range corner, but that has not been measured, and with it the extrapolation of section 4 is untested.

The origin of the non-circularity is not attributed here. The residual is smooth and antisymmetric-plus-peak in shape, which is consistent with an unmodelled series element, with a frequency-dependent error in the amplitude or phase calibration, or with the constant-baseline removal applied before publication. Distinguishing these requires a measurement this document does not contain.

The reference is itself a fit, not ground truth. Where the model residual is 4 % of the radius, calling any one of these circles correct to better than a few percent is not supported by the data.

References

Taubin, G. (1991). Estimation of planar curves, surfaces and nonplanar space curves defined by implicit equations, with applications to edge and range image segmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence 13(11), 1115-1138. The algebraic estimator used as the panel's fit and as the initial guess of the geometric one.

Chernov, N. and Lesort, C. (2005). Least squares fitting of circles. Journal of Mathematical Imaging and Vision 23, 239-252. The geometric, orthogonal-distance formulation and its relation to the algebraic estimators.